With the 5 V supply complete, our next concern is the +12 V and −12 V supplies. Other devices within the 7800 family of regulators will satisfy the requirements. The 7812 and a 7912 are 1 A positive and negative regulators respectively; they exhibit the same protection characteristics as the 78H05. Figures 1.14 and 1.15 outline the exact specifications. Because we are dealing with much lower currents than the +5 V supply, there is considerably less concern over voltage losses through connecting cables, and it is unnecessary to add trim adjustment circuitry. Figure 1.16 is the finished schematic of the ZAP power supply. Additional regulator circuit diagrams (figures 1.17a, b, c and d) are included to demonstrate how the 7800 series of regulators can be used in our application. Are we finished yet? Of course not. Close examination of figure 1.16 shows two items not discussed previously: heat sinks and overvoltage protection. These two subjects and a short discussion of the importance of correct layout complete the chapter.

μA7812

ELECTRICAL CHARACTERISTICS: V_{IN} = 19 V, I_{OUT} = 500 mA, −55°C < T_J < 150°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF, unless otherwise specified.

CHARACTERISTICS		CONDITIONS		MIN	TYP	MAX	UNITS
Output Voltage		T_J = 25°C		11.5	12.0	12.5	V
Line Regulation		T_J = 25°C	14.5 V < V_{IN} < 30 V		10	120	mV
			16 V < V_{IN} < 22 V		3.0	60	mV
Load Regulation		T_J = 25°C	5 mA < I_{OUT} < 1.5 A		12	120	mV
			250 mA < I_{OUT} < 750 mA		4.0	60	mV
Output Voltage		15.5 V < V_{IN} < 27 V 5 mA < I_{OUT} < 1.0 A P < 15 W		11.4		12.6	V
Quiescent Current		T_J = 25°C			4.3	6.0	mA
Quiescent Current Change	with line	15 V < V_{IN} < 30 V				0.8	mA
	with load	5 mA < I_{OUT} < 1.0 A				0.5	mA
Output Noise Voltage		T_A = 25°C, 10 Hz < f < 100 kHz			8	40	μV/V_{OUT}
Ripple Rejection		f = 120 Hz, 15 V < V_{IN} < 25 V		61	71		dB
Dropout Voltage		I_{OUT} = 1.0 A, T_J = 25°C			2.0	2.5	V
Output Resistance		f = 1 kHz			18		mΩ
Short Circuit Current		T_J = 25°C, V_{IN} = 35 V			0.75	1.2	A
Peak Output Current		T_J = 25°C		1.3	2.2	3.3	A
Average Temperature Coefficient of Output Voltage		I_{OUT} = 5 mA	−55°C < T_J < +25°C			0.4	mV/°C/ V_{OUT}
			+25°C < T_J < +150°C			0.3	

Figure 1.14 *Electrical characteristics of the μA7812 voltage regulator.*

μA7912

ELECTRICAL CHARACTERISTICS: V_{IN} = −19 V, I_{OUT} = 500 mA, C_{IN} = 2 μF, C_{OUT} = 1 μF, −55°C < T_J < 150°C, unless otherwise specified.

CHARACTERISTICS		CONDITIONS		MIN	TYP	MAX	UNITS
Output Voltage		T_J = 25°C		−11.5	−12.0	−12.5	V
Line Regulation		T_J = 25°C	−14.5 V < V_{IN} < −30 V		10	120	mV
			−16 V < V_{IN} < −22 V		3.0	60	mV
Load Regulation		T_J = 25°C	5 mA < I_{OUT} < 1.5 A		12	120	mV
			250 mA < I_{OUT} < 750 mA		4.0	60	mV
Output Voltage		−15.5 V < V_{IN} < −27 V 5 mA < I_{OUT} < 1.0 A P < 15 W		−11.4		−12.6	V
Quiescent Current		T_J = 25°C			1.5	3.0	mA
Quiescent Current Change	with line	−15 V < V_{IN} < −30 V				1.0	mA
	with load	5 mA < I_{OUT} < 1.0 A				0.5	mA
Output Noise Voltage		T_A = 25°C, 10 Hz < f < 100 kHz			25	80	μV/V_{OUT}
Ripple Rejection		f = 120 Hz, −15 V < V_{IN} < −25 V		54	60		dB
Dropout Voltage		I_{OUT} = 1.0 A, T_J = 25°C			1.1	2.3	V
Peak Output Current		T_J = 25°C		1.3	2.1	3.3	A
Average Temperature Coefficient of Output Voltage		I_{OUT} = 5 mA, −55°C < T_J < 150°C				0.3	mV/°C/ V_{OUT}
Short Circuit Current		V_{IN} = −35 V, T_J = 25°C				1.2	A

Figure 1.15 *Electrical characteristics of the μA7912 voltage regulator.*

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